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Ecology as Organised Persistence Across Scales

Canonical Article | [Ecology](#) | Created: 28 May 2026 | Revised: 9 September 2026

Anchor

 **Canonically locked** since 28 May 2026 · Revised 9 September 2026.

Abstract

This article develops the APS account of ecology as materially realised organism–environment organisation relevant to biological persistence. Ecology is not treated as an external environmental backdrop surrounding independently complete organisms, nor as persistence produced by interacting scales. Living systems depend upon resources, constraints, developmental conditions, ecological relations, microbial communities, and changing environments that can contribute directly to their continued viability. APS therefore asks which ecological relations make an explanatory difference to a specified biological target and what spatial and temporal extent is required to explain them. Some ecological explanations require relations extending beyond organism boundaries and across broad extents; others may be adequately bounded more narrowly. Scale identifies that explanatory extent but does not perform the ecological interaction, coupling, constraint, or causal work.

Key Points

- Ecology concerns materially realised organism–environment relations relevant to organised persistence.
- Some conditions of organismal persistence are realised through relations extending beyond organism boundaries.
- Distributed persistence does not mean persistence distributed among scales.
- Scale identifies the spatial and temporal extent required for an ecological explanation; scales do not interact or cause.

- A broader ecological extent is warranted only when additional materially realised relations make an explanatory difference to the explanandum.
- Organism–environment coupling is a material relation and must not be confused with coupling among scales.
- Perturbation can reveal ecological dependencies that require expansion of an initially bounded explanation.
- Resilience and adaptation concern material reorganisation under changing ecological conditions.
- Ecological significance is viability-relative and does not by itself establish cognition.
- Broader or multiextent ecological explanation is a diagnostic outcome, not a universal premise.

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Introduction

Where this article fits: *The Ecological Organisation of Life* introduces the ecological organisation of biological persistence. This article asks a more specific explanatory question: when does understanding persistence require organism–environment relations extending beyond a narrowly bounded biological system, and how should the required explanatory extent be identified?

Ecology is often introduced as the study of organisms and their environments. While useful as a starting point, this description leaves open a deeper question. Why do organism–environment relations matter so fundamentally to life in the first place?

APS approaches this question through organised persistence. Living systems do not persist independently of the conditions with which their organisation is materially related. Energy sources, nutrients, atmospheric and chemical conditions, developmental environments, microbial communities, ecological relationships, behavioural opportunities, and other environmental conditions can contribute directly to continued viability. Ecology therefore concerns more than a backdrop surrounding otherwise complete organisms. It investigates relations through which the conditions of biological persistence can be maintained, altered, or lost.

This does not mean that ecology is inherently organised through interacting scales. The relevant explanatory question is more discriminating. For a specified ecological explanandum, which materially realised processes,

constraints, interactions, and organisational relations make an explanatory difference, and across what spatial and temporal extent must they be considered?

Some ecological explanations require relations extending substantially beyond organism boundaries or across long temporal intervals. Others may be adequately explained within a more bounded domain. APS does not decide between these possibilities in advance.

Ecological explanation therefore identifies the materially realised organism–environment relations relevant to persistence; **Scale identifies the spatial and temporal extent required to explain them.**

APS Box — What Biological Explanation Explains

Within APS, living systems are understood as viability-oriented, constraint-closed organisations whose activity maintains and re-establishes conditions required for continued functioning.

Agency, Process, and Scale are complementary analytic projections of this one organisation. **Agency** concerns what living systems do. **Process** concerns how organisational continuity is maintained despite change. **Scale** concerns where persistence is organised across spatial and temporal extents, with the required explanatory extent determined by the materially realised relations relevant to the explanandum.

These projections help formulate different questions about living organisation without determining in advance either the explanandum or what an adequate explanation must contain. APS may ask how viability is maintained, how constraints are mutually sustained, how continuity persists through transformation, or how materially implemented relations across spatial and temporal extents alter local conditions.

Other APS concepts identify more specific explanatory relations. Biological Evaluation concerns the process through which agency generates significance. Function concerns present-tense viability-relative organisational contribution. Purpose concerns viability-oriented organisation without requiring intention or future-directed causation. Cognition requires an organisation of biological significance that goes beyond immediate evaluation or fixed regulation. Evolution concerns historical change and remains explanatorily distinct from the present organisation of living systems.

APS therefore does not treat all biological phenomena as expressions of organised persistence or require every explanation to employ its full conceptual vocabulary. Organised persistence denotes the continuity of living organisation through material and structural transformation. Where APS and another framework address the same explanatory target, any claimed explanatory gain requires target-matched comparison with the strongest relevant alternative.

The central ecological question becomes:

How do materially realised organism–environment relations contribute to the persistence relevant to a specified ecological explanandum, and what explanatory extent is required to account for that contribution?

APS Box — How APS Concepts Relate

APS does not organise inquiry around a single privileged explanatory principle or level. It uses a related set of concepts to distinguish different questions concerning viability-oriented living organisation and organised persistence.

- **Viability** identifies conditions relevant to whether living organisation can continue functioning and persist.
- **Persistence** concerns the continuation and re-establishment of organised activity across time.
- **Constraint closure** identifies an APS organisational commitment concerning the reciprocal maintenance of constraints involved in living organisation.

- **Coupling** describes materially realised relations through which systems, processes, or conditions influence one another.
- **Biological agency** is viability-oriented organisational activity.
- **Semiosis** belongs to an APS research pathway investigating how viability-relevant differences may acquire biological significance; its stronger dependency claims require independent assessment.
- **Resolution** identifies the explanatory granularity at which organisation is described or analysed.

These concepts do not have identical functions or evidential status, and their inclusion within APS does not make each concept necessary for every biological explanation.

Agency, Process, and Scale have a more specific architectural role: they are complementary, target-sensitive analytic projections of the same living organisation rather than components, levels, stages, or independent causes.

The concepts collected here therefore provide different ways of formulating and investigating APS questions about living organisation. Their conceptual integration within the framework does not by itself establish additional explanatory power, biological truth, or superiority over independently established explanations.

Organisms Are Ecologically Constituted

Traditional biological explanation can encourage a picture in which the organism is treated as the primary biological unit while the environment appears only as an external source of inputs, disturbances, or selection pressures. That separation becomes inadequate where conditions beyond an organism's boundary contribute materially to the organisation through which it remains viable.

Every organism depends upon environmental relations of some kind. Energy must be acquired, materials exchanged, appropriate physicochemical conditions encountered or maintained, and activity regulated relative to circumstances beyond the organism itself. For many organisms, microbial partnerships, developmental environments, social relations, constructed niches, or other ecological dependencies can also make a material difference to persistence.

Such relations are not made explanatory merely because they occur outside an organism. Their relevance must be established relative to the target being explained. A nutritional dependency may be indispensable to one explanandum but irrelevant to another; a microbial relation may be constitutive of a developmental outcome without being required for every explanation of the organism concerned.

APS therefore permits a strong but bounded ecological claim: **some conditions of organismal persistence are materially realised through relations extending beyond organism boundaries.**

This does not dissolve organisms into environments. Organisms remain materially organised living systems whose activities can modify the conditions with which they interact. Environments likewise affect organisms through specific physical, chemical, biological, and ecological relations. Where reciprocal dependence matters to the explanandum, explanation should identify those relations rather than assign the causal work to a wider scale.

APS Box — Failure Reveals Organisation

In APS, failure is not merely the absence of successful functioning. It can provide evidence about the dependencies and limits of viability-oriented organisation.

Living systems maintain and re-establish conditions required for continued functioning. Because those conditions are vulnerable to disruption, malfunction, breakdown, and loss of viability can make otherwise unobtrusive organisational dependencies experimentally accessible. Failure may reveal what particular activities

or relations contribute, where regulatory limits lie, and under what conditions functioning can or cannot be restored.

Repair, compensation, reorganisation, and recovery can likewise provide evidence about the activities, constraints, and dependencies involved in maintaining living organisation. These responses remain materially and environmentally implemented; they do not imply that living systems are organisationally independent of external conditions.

Failure is therefore diagnostically useful where contrasts between successful functioning, impairment, recovery, and irreversible breakdown identify relations that make a relevant difference to viability. Malfunction can help identify present-tense viability-relative organisational contributions, while breakdown can reveal dependencies required for continued organisation. Death marks the irreversible loss of the viability-oriented, constraint-closed organisation constitutive of Life.

Vulnerability is consequently important to the study of living organisation because viability can be maintained, impaired, restored, or lost under different conditions. Failure is one investigative route through which these differences and their organisational dependencies can become tractable.

Key Point: Failure, repair, and recovery can reveal dependencies relevant to living organisation, but disruption does not by itself establish an APS explanation. Any claimed explanatory contribution must be assessed relative to the explanandum and relevant comparator.

Ecology as Distributed Organised Persistence

Ecological organisation becomes especially important where the conditions relevant to persistence cannot be adequately represented as wholly internal to one bounded biological system.

Predation, symbiosis, microbiome relations, nutrient cycling, developmental scaffolding, ecological signalling, niche construction, and multispecies interactions provide different examples of relations that may contribute to biological persistence. They are not unified by physical similarity, and they should not automatically be treated as instances of one universal ecological mechanism. Their common relevance is methodological: each can become part of an explanation when it makes a material difference to the persistence being explained.

APS uses **distributed organised persistence** in this bounded sense. Persistence is distributed when materially realised conditions relevant to a specified persistence explanandum occur across relations that cannot be adequately represented as wholly internal to one bounded biological system.

Distribution therefore does not mean that persistence is distributed among scales or propagated from one scale to another. The relevant processes, constraints, interactions, and organisational relations do the biological work. Scale identifies how far across space and time the explanation must extend to include them.

This distinction allows ecological explanations to be broad where the biology requires breadth without treating breadth itself as explanatory.

The question is therefore not:

Across how many scales is persistence distributed?

but:

Which ecological relations make a difference to the persistence being explained, and what spatial and temporal extent is required to include them?

Ecology and Viability

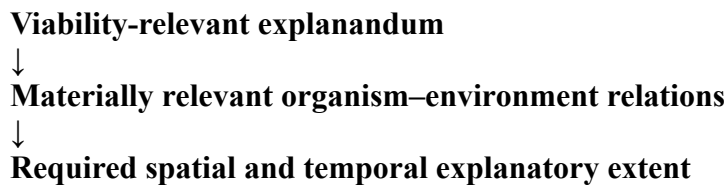
Ecological conditions matter biologically because some of them alter the prospects for continued viable organisation.

Changes in resource availability, temperature, habitat conditions, predation, toxicity, microbial relations, developmental circumstances, or other environmental variables can support, disrupt, or transform the organisation through which living systems persist. Their biological significance therefore derives not simply from being external circumstances but from their relation to viability.

This helps explain why ecological conditions matter to biological agency. Organisms act relative to conditions that bear differently upon their continued functioning. Resources can support persistence; hazards can threaten it; environmental opportunities can enable activities through which viable organisation is maintained or re-established.

The relevant viability is not itself distributed among scales. Rather, explaining a particular viability outcome may require environmental relations extending beyond a narrowly specified boundary. Whether that broader extent is necessary must be established from the explanandum.

The ecological explanatory sequence is therefore:



This sequence prevents explanatory extent from being mistaken for the material organisation being explained.

Ecological Explanation and Explanatory Extent

Ecological phenomena vary greatly in the spatial and temporal extent required for their explanation.

A local resource interaction may sometimes be explained adequately within a narrowly bounded organism–environment relation. Other targets may require developmental history, microbial communities, population processes, habitat transformation, climatic conditions, or historical ecological relations extending substantially beyond that boundary.

APS does not assume in advance that the broader explanation is superior.

The controlling test is:

A broader spatial or temporal extent is warranted in ecological explanation only when materially realised relations beyond a narrower boundary make a demonstrable explanatory difference to the ecological explanandum.

This yields two equally legitimate explanatory outcomes:

Bounded ecological explanation

or:

Broader / multiextent ecological explanation

A broader explanation earns its place by explanatory contribution, not by including more spatial domains, longer periods, or more biological processes. Conversely, a bounded explanation should not be rejected merely because wider ecological relations exist.

Scale therefore asks **where the persistence relevant to the ecological explanandum is organised across spatial and temporal extents**. It does not coordinate those extents or cause relations between them.

Broader ecological extent is a diagnostic outcome, not a universal premise.

Organism–Environment Coupling

Within APS, organism–environment coupling refers to materially realised relations through which living activity and environmental conditions affect one another.

Living systems regulate exchanges with their environments. They acquire resources, avoid threats, exploit opportunities, modify habitats, alter local chemical or physical conditions, and reorganise activity relative to changing circumstances. Such activities can contribute directly to the maintenance or re-establishment of viable organisation.

Environments are therefore not always passive backgrounds. Their conditions can participate materially in developmental, physiological, behavioural, adaptive, or other biological processes.

But processes and systems may be coupled; scales are not.

Where organism–environment coupling extends beyond a local boundary or over a longer temporal interval, Scale identifies the extent required to explain that coupling. It does not perform the coupling or propagate continuity between extents.

Ecological affordances provide one example of why the relation matters. A resource for one organism may be irrelevant to another; a condition compatible with persistence for one form of life may threaten another. The ecological significance of such conditions depends upon the relation between environmental difference and viability-oriented organisation.

APS consequently treats organism–environment coupling as materially real where the relevant interactions and dependencies are materially implemented. The explanatory task is to identify those relations and determine whether they require a bounded or broader ecological account.

Ecology and Development

Development occurs through continuing engagement with conditions that can contribute materially to developmental organisation.

Nutrition, microbial ecologies, temperature, chemical conditions, behavioural environments, social relations, developmental scaffolding, and ecological perturbations can influence developmental trajectories. Such influences cannot be understood simply by labelling development and ecology as different levels.

The relevant question is whether a particular developmental explanandum depends upon materially realised relations extending beyond the boundary initially chosen for explanation.

Molecular and cellular processes may be sufficient for some developmental questions. Other developmental outcomes may require physiological, behavioural, microbial, ecological, or historical relations. The appropriate explanatory extent must therefore be established case by case.

APS expresses this as follows:

Development can depend upon materially realised environmental conditions, resources, interactions, and constraints. Where such relations make a difference to a developmental explanandum, their spatial and temporal extent belongs within the explanation.

Developing organisms can also alter their environments through activity, resource use, signalling, habitat modification, and social or ecological interaction. Where this reciprocity matters, explanation should identify the material relations through which it occurs.

Development therefore illustrates how an ecological explanation can legitimately extend beyond an organism without making multiextent explanation a universal requirement.

Ecology and Perturbation

Ecological systems and organism–environment relations are continually exposed to changing conditions.

Climatic fluctuations, habitat disruption, resource variation, predation, species interactions, pollution, anthropogenic transformation, and other perturbations can alter conditions relevant to persistence. Such changes often reveal dependencies that are difficult to identify during relatively stable functioning.

APS therefore treats perturbation as diagnostically valuable. A disturbance can reveal which resources, constraints, interactions, redundancies, or environmental relations actually make a difference to the target being explained.

A microbial disturbance may alter development; a developmental disruption may change behaviour; altered behaviour may modify ecological relations; ecological transformation may affect later evolutionary possibilities. These sequences should not be described as scales acting upon one another. They are candidate chains of materially realised dependency whose explanatory relevance must be demonstrated.

Perturbation can therefore alter the justified boundary of an explanation:

Where perturbation demonstrates that relations beyond an initially bounded domain make a difference to the target, the warranted explanatory extent expands accordingly.

This makes perturbation a method for discovering explanatory extent rather than evidence that ecological organisation must always be multiextent.

APS Box — Perturbation Reveals Organisation

In APS, perturbation can be diagnostically useful because disruption may make organisational dependencies visible that are difficult to distinguish under relatively stable conditions.

A system observed only during successful functioning may not reveal which relations contribute to maintaining that functioning. Perturbation can establish contrasts through which regulation, compensation, repair, reorganisation, dependence, and failure become empirically accessible.

Under disruption, a system may degrade, remain dependent upon external support, compensate partially, or reorganise its activity in ways that maintain or re-establish conditions required for continued functioning. Comparing these responses can reveal dependencies, regulatory capacities, limits of viability, and relations involved in maintaining living organisation.

Perturbation is therefore a useful investigative strategy rather than a universally privileged diagnostic method. Not every organisational relation requires perturbation to be identified, and disruption does not by itself establish an APS explanation.

Where perturbation reveals that particular activities, constraints, or relations make a relevant difference to viability, those contrasts can provide evidence for an organisational explanation. Constraint closure, where relevant, concerns the mutual maintenance of constraints involved in sustaining living organisation rather than independence from environmental or material support.

Key Point: Perturbation can make organisational dependencies experimentally tractable by revealing how a living system responds when conditions affecting viability are altered. Whether those dependencies provide explanatory gain must still be established relative to the explanandum and relevant comparator.

Ecology and Resilience

Ecological persistence frequently requires organisation to remain viable under changing conditions.

Resilience therefore involves more than resistance to disturbance. In relevant cases it concerns the capacity of materially organised relations to preserve or re-establish continuity through reorganisation when conditions change.

Different mechanisms may contribute: biodiversity, behavioural flexibility, developmental plasticity, altered trophic relations, redundancy, migration, physiological adjustment, habitat modification, or ecosystem transformation. These mechanisms should not be treated as interchangeable. Their explanatory relevance depends upon the particular resilience target.

Some resilience explananda may be adequately bounded. Others may depend upon relations extending beyond a local system or over longer periods. The explanatory extent should follow the demonstrated dependencies.

APS therefore treats ecological resilience as **continuity-preserving or continuity-restoring reorganisation under perturbation**, where that description is appropriate to the target. Processes and relations reorganise; scales do not.

Ecology and Adaptation

Adaptation occurs in relation to environmental conditions relevant to biological persistence and reproduction.

Ecological relations can influence developmental trajectories, behaviour, selection pressures, and the possibilities available to organisms. Organisms can in turn modify environmental conditions through niche construction, migration, cooperation, resource use, environmental engineering, and other activities.

This reciprocity is materially realised. It should not be redescribed as reciprocal influence among scales.

An adaptive explanandum may require developmental, behavioural, ecological, or evolutionary relations, but their inclusion must be justified by their explanatory contribution. A broader account is warranted where excluding those relations leaves a demonstrable explanatory deficit.

APS therefore treats adaptation as occurring within organism–environment organisation without assuming that every adaptive explanation must extend across the same spatial or temporal domains.

Ecological conditions can change; organisms can respond to and modify those conditions; and the resulting relations can alter future possibilities for persistence. The explanatory task is to identify the material dependencies and their relevant temporal extent.

Ecology and Semiosis

Ecology is not only a matter of material exchange. Living systems also encounter environmental differences that matter unequally to their continued functioning.

Food, predators, toxins, shelter, mates, climatic conditions, competitors, gradients, and signals can have different consequences for viable organisation. Their significance is relational: a condition matters biologically in relation to the organisation for which it bears upon viability.

Within APS, **Biological Evaluation is the process through which agency generates significance**. Ecological circumstances can participate materially in the relation being evaluated, but neither environments nor scales generate significance simply by occupying a wider explanatory domain.

Signals, gradients, cues, and other environmental differences can therefore become biologically significant through their relation to viability-oriented activity. A chemical trace may be relevant to resource acquisition; a sound may indicate danger; an environmental change may alter the conditions under which persistence can be maintained.

The explanatory extent required for semiosis depends on the target. Some cases may be explained through relatively local organism–environment relations. Others may require developmental history or broader ecological conditions. That extent must be demonstrated rather than assumed.

APS consequently treats ecological significance as relationally real without distributing significance among scales or attributing evaluation indiscriminately to ecological systems.

Ecology and Cognition

Ecological organisation provides important conditions within which cognition can operate, but ecological responsiveness alone does not establish cognition.

Perception, learning, behavioural coordination, and other cognitive processes, where present, occur through engagement with viability-relevant circumstances. Ecological conditions can structure the possibilities organisms encounter and can contribute to the significance that cognitive activity integrates across time.

But organism–environment coupling, adaptive response, signalling, or ecological significance does not by itself cross the APS threshold into cognition.

APS therefore preserves a stronger distinction. Biological agency becomes cognitive when integrated biological significance modulates activity across a temporal field of viability-relevant possibilities in ways not exhausted by immediate or fixed regulation.

The ecological contribution is consequently bounded. Ecology can help explain the conditions, relations, and significance within which cognition operates. It does not independently establish the cognitive threshold.

Where a cognitive explanandum requires environmental relations extending beyond a narrowly bounded organism, those relations belong within the explanation. Where they do not make an additional explanatory difference, broader ecological extent should not be added merely because cognition occurs in an environment.

Ecology and Constraint Organisation

Ecological organisation includes materially implemented constraints that shape the possibilities available to biological activity.

Resource distributions, physical boundaries, chemical conditions, trophic relations, habitat structure, developmental dependencies, competition, symbiosis, and other ecological relations can constrain what living systems can do. Constraints need not merely suppress activity; they can participate in organising the conditions under which particular activities and forms of persistence remain possible.

The explanatory question is not whether constraints operate at one level and influence another. It is:

Which materially implemented constraint alters which process or condition, and what consequence does that relation have for the explanandum?

Only after those relations are identified does Scale enter the analysis by asking what spatial and temporal extent is required to explain them.

Ecological constraint organisation can therefore extend beyond organism boundaries without becoming constraint by Scale. Networks of ecological dependencies may require broad explanation where their constituent relations demonstrably contribute to the target.

APS consequently approaches ecological constraint organisation through:

materially realised constraints and dependencies relevant to organised persistence, considered across whatever spatial and temporal extent their explanation requires.

Ecology, Evolution, and Historical Continuity

Ecology and evolution are deeply related, but their relation should not be represented as interaction between different scales.

Evolutionary transformations occur within environmental conditions and ecological relations that can affect survival, reproduction, development, and other processes relevant to evolutionary change. Evolutionary transformations can in turn alter ecological relations by producing new forms, activities, dependencies, interactions, and environmental modifications.

The relation is therefore materially and historically reciprocal where the relevant processes support that conclusion.

Ecological conditions can contribute to evolutionary explananda, while evolutionary transformations can alter later ecological conditions. Whether a particular explanation requires both domains depends upon the target.

Evolution concerns historical transformation across lineages. Ecology concerns organism–environment relations and other ecological organisation. Their explanatory domains can overlap without constituting interacting ontological levels or scales.

Where historical ecological relations are necessary to explain an evolutionary target, the required temporal extent expands accordingly. Where an ecological target can be explained without reconstructing longer evolutionary history, that additional extent need not be imposed.

APS therefore treats ecology and evolution as distinct but potentially overlapping explanatory domains within the study of organised persistence.

Beyond Reductionism and Holism

Ecological explanation is sometimes framed as a choice between reduction to local mechanisms and appeal to larger ecological wholes.

APS does not resolve this dispute by declaring either explanatory breadth or explanatory narrowness superior in advance.

A narrowly bounded explanation may be entirely adequate when the materially relevant relations required for the explanandum lie within that boundary. It becomes insufficient when relations excluded by the boundary make a demonstrable explanatory difference.

A broader ecological explanation faces the reciprocal requirement. Including more organisms, processes, spatial domains, or temporal history does not itself produce explanatory gain. The additional relations must contribute to explaining the target.

The relevant methodological principle is therefore:

The adequacy of a narrower or broader ecological explanation must be assessed relative to the explanandum and an appropriate comparator. A narrower explanation is insufficient only when materially relevant relations excluded by its boundary provide demonstrable explanatory gain. A broader explanation is not superior merely because it includes more relations.

This approach preserves mechanistic specificity while permitting explanations to extend beyond local mechanisms where the biology warrants that extension.

APS therefore replaces neither reductionism with holism nor local explanation with multiextent explanation. It asks what organisation actually explains the target and how far across space and time that explanation must extend.

Why Ecology Matters in APS

Ecology matters to APS because living systems do not maintain viable organisation independently of their material circumstances.

Energy sources, resources, developmental conditions, microbial relations, ecological communities, physical environments, and changing external conditions can contribute directly to the persistence of living systems. Organism–environment organisation can therefore be constitutive rather than merely contextual for particular biological explananda.

Ecology also provides a strong test of explanatory boundaries. Some biological questions can be answered within relatively bounded systems. Others expose dependencies extending beyond organism boundaries, across communities or environments, or through longer temporal relations.

The significance of Scale is methodological. It allows APS to ask where the persistence relevant to the explanandum is organised without converting spatial or temporal extent into an additional causal entity.

Ecology consequently demonstrates why explanatory boundaries should be tested rather than assumed. A local boundary may prove sufficient. It may instead fail when perturbation, comparison, or dependency analysis reveals materially relevant relations outside it.

Ecological explanation is therefore especially important to APS because it can show, rather than presuppose, when organised persistence requires a broader explanatory extent.

Conclusion

APS reconstructs ecology around materially realised organism–environment relations relevant to organised persistence.

Living systems depend upon conditions that can extend beyond their boundaries: resources, constraints, developmental circumstances, microbial relations, ecological interactions, meaningful environmental differences, and historically changing environments. Ecology investigates how such relations contribute to biological persistence, transformation, vulnerability, and recovery.

These relations do not establish that ecology is inherently multiscale. Nor do they imply that scales interact, coordinate processes, transmit constraints, or propagate persistence.

Instead, ecological explanation begins with an explanandum. It identifies the materially realised processes, constraints, interactions, and organisational relations that make a difference to that target. Scale then identifies

the spatial and temporal extent required to include them.

Some explanations will remain bounded. Others will require broader or multiextent accounts. The distinction must be earned by explanatory evidence.

Organism–environment coupling explains material relations between living systems and their conditions. Perturbation can expose dependencies that alter the justified boundary of explanation. Resilience concerns continuity-preserving or continuity-restoring reorganisation. Adaptation concerns biological change relative to ecological conditions. Semiosis concerns viability-relative significance. Cognition, where present, organises integrated biological significance across a temporal field of possibilities. Evolution concerns historical transformation across lineages that can intersect materially with ecological organisation.

Taken together, these themes establish the ecological contribution to APS without making ecological breadth an explanatory premise.

Key Point. Ecological relations can be materially constitutive of the conditions through which living systems persist, and those relations may extend beyond organism boundaries across substantial spatial or temporal extents. The required extent is determined by the explanandum and the materially realised relations that make a difference to it. Broader ecological explanation is therefore a possible diagnostic result, not a universal premise. Scale identifies where the relevant persistence must be explained; it does not perform the ecological interaction, coupling, constraint, or causal work.

[Explanatory Architecture](#)

[Central Question](#)

Which materially realised organism–environment relations contribute to the persistence relevant to an ecological explanandum, and what spatial and temporal extent is required to explain that contribution?

[Architectural Role](#)

This article establishes the ecological Anchor for analysing organised persistence beyond an automatically organism-bounded explanatory frame. It shows how ecological relations can enter biological explanation without treating environments as passive backgrounds or Scale as a material component of ecological organisation.

[Builds On](#)

The article builds on APS accounts of life as viability-oriented, constraint-closed organisation; biological agency as viability-oriented organisational activity; organised persistence; organism–environment coupling; viability; constraint; Process; and Scale.

It also depends on the distinction between materially realised biological organisation and the analytic projections through which that organisation is explained.

[Develops Into](#)

The architecture developed here supports more specific APS analyses of organism–environment coupling, ecological resilience, adaptation, development, semiosis, cognition, and evolutionary transformation where ecological relations make an explanatory difference.

It also provides a methodological basis for asking when ecological explanations require broader spatial or temporal extent and when a more bounded explanation is sufficient.

[Related Explanatory Questions](#)

Related questions include:

- When is an environmental condition constitutive of a biological explanandum rather than merely contextual?
- How can perturbation reveal ecological dependencies that stable observation conceals?
- Which materially implemented constraints connect organismal activity with ecological conditions?
- When does ecological resilience require explanation beyond a local boundary?
- How do ecological circumstances acquire viability-relative significance?
- When do developmental or evolutionary histories make an additional explanatory difference to an ecological target?
- What evidence warrants expanding the spatial or temporal extent of an ecological explanation?

[Position within APS](#)

Ecology does not constitute a fourth APS projection. **Agency, Process, and Scale are analytic projections of one living organisation.**

Agency asks what living organisation does. Process asks how continuity is maintained despite change. Scale asks where persistence is organised across spatial and temporal extents.

Ecological explanation concerns particular biological explananda and the materially realised organism–environment relations relevant to them. It may draw upon one or more APS projections according to the explanatory target, but it does not establish that every ecological explanation requires all three or that Scale is automatically indispensable.

[Corpus Significance](#)

This article establishes a discriminating ecological application of APS. Its central contribution is not the claim that ecology is inherently multiscale, but a method for determining when relations beyond an initially bounded system provide explanatory gain.

Ecology therefore becomes an important test case for APS comparative explanatory methodology: broader explanation is warranted when additional materially realised ecological relations make a difference to the target, and not simply because wider relations can be described.

[Where to Go Next](#)

- **The Ecological Organisation of Life**
- **Organism–Environment Coupling**
- **Ecological Resilience**
- **Niche Construction**
- **Adaptation — How Living Systems Sustain Themselves Through Change**
- **Semiosis — How Differences Come to Matter in Living Systems**
- **Cognition — Where Does It Belong in Biology?**
- **Constraint — How Organisation Becomes Persistence**
- **Biological Agency — The Activity of Organised Persistence**

Together these articles explore how ecological relations contribute to organised persistence across development, adaptation, resilience, cognition, evolution, and other biological explananda.

See Also

- [Ecology](#)
- [Ecological Organisation](#)
- [Organism–Environment Coupling](#)
- [Persistence](#)
- [Viability](#)
- [Biological Agency](#)
- [Adaptation](#)
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- [Process](#)
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- [Coupling](#)
- [Constraint](#)
- [Constraint Closure](#)
- [Development](#)

Related Articles

- [The Ecological Organisation of Life](#)
- [Organism–Environment Coupling — Distributed Persistence Across Living Systems](#)
- [Biological Individuality — Organised Persistence Through Time](#)
- [Biological Agency as the Activity of Self-Maintenance](#)
- [Persistence — Organised Continuity Through Time](#)
- [Viability — The Organising Principle of Biological Persistence](#)
- [Temporal Organisation and Organised Persistence](#)
- [Adaptation — How Living Systems Sustain Themselves Through Change](#)
- [Resilience, Ecology, and Continuity](#)
- [Semiosis — How Differences Come to Matter in Living Systems](#)
- [Cognition — Where Does It Belong in Biology?](#)
- [Constraint — How Organisation Becomes Persistence](#)
- [The Explanatory Geometry of Biology — How APS Organises Biological Explanation](#)
- [Diagnosis as Continuity Analysis](#)
- [What Is Evolution in APS?](#)

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